

Technical Requirement Document (TRD) — Stitch AI Data Suite

Document Version: 1.0.0
System Target: Python 3.11 / Streamlit Runtime Layer
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1. System Architecture & Context Ingestion

Stitch operates as a client-facing web instance where state isolation occurs entirely inside isolated memory boundaries. The platform is structured defensively to split data interactions into native deterministic code execution and asynchronous language model telemetry context generation.

2. Core Project Dependencies

The ecosystem depends on strict version parameters across several framework packages to maintain multi-threaded stability and high performance:

Dependency Component	Minimum Version Requirement	Architectural Operational Role
streamlit	>= 1.32.0	Core presentation engine, multi-threaded worker rendering, and page routing.
pandas	>= 2.0.0	Dataframe file structures, algorithmic sanitization, and descriptive metric tracking.
scikit-learn	>= 1.4.0	Automated feature modeling, variable transformations, cross-validation scoring.

Dependency Component	Minimum Version Requirement	Architectural Operational Role
google-generativeai	>= 0.5.0	Secure API calls mapping requests out to Google Gemini models.
plotly	>= 5.20.0	Interactive, web-optimized asset canvas rendering for metrics.
xgboost	>= 2.0.0	High-speed tree gradient boosting for model tracking.
reportlab	>= 4.1.0	Deterministic generation of executive business report PDFs.

3. Critical Architectural Bug Fixes & Preload Strategies

To solve multi-threaded runtime errors in Python (such as the `RuntimeError: can't register atexit` after shutdown exception), the application must force thread process pool configurations to register on the primary thread during initialization. The following sequential script must be explicitly placed at the root entry point of `app.py` before executing standard stream calls:

```
import concurrent.futures.thread
import concurrent.futures.process
import sklearn # Pre-loads thread execution structures inside the core runtime layer
import streamlit as st
```

4. Module-Level Technical Specifications

Each sub-module operates within strict execution boundaries to prevent runtime exceptions:

- **Session-State Management:** Raw datasets are stored as explicit keys (`df_raw` and `df_cleaned`). To prevent Pandas truth value evaluation ambiguities, files must never be

tested using implicit boolean operators (e.g., A or B). Instead, they are resolved using explicit None conditional testing.

- **Hybrid Conversational Assistant Model:** Natural language strings are checked against compiled regular expressions to process basic statistical requests instantly. If no pattern matches, the system extracts a secure text context string containing only the dataset's structural schema, descriptive statistics, and up to five sample rows, avoiding any full-data leakage.
- **AutoML Cross-Validation Workflow:** Columns are parsed through targeted label encoders and missing rows are filled using median calculations. Models compete over 3-fold cross-validation metrics. Classification problems are scored using accuracy parameters, while regression configurations default to R^2 scoring targets.

5. Security, Isolation, & Model Configuration Matrix

The intelligence engine utilizes Google's modern gemini-2.5-flash model endpoint to prevent service deprecation issues. For maximum data protection, full datasets are never serialized or sent across the network. Built-in PII validation rules run local token scanning checks across active data tables to mask columns before metadata summaries are compiled into JSON format for API consumption.